

Amelioration of degraded soils under red pine plantations on the Oak Ridges Moraine, Ontario

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McPherson, T. S. and Timmer, V. R. 2002. **Amelioration of degraded soils under red pine plantations on the Oak Ridges Moraine, Ontario.** Can. J. Soil Sci. **82**: 375–388. Soil degradation and subsequent amelioration were studied on soil chronosequences of old-growth forest, abandoned fields, and young and mature conifer plantations on the Oak Ridges Moraine, an environmentally vulnerable landform near Toronto threatened by encroaching urban development. The chronosequences reflect a history of pre-settlement deforestation, exploitive pioneer agriculture and ensuing land abandonment that led to soil fallowing and/or wind erosion in the 1920s followed by soil stabilization after extensive planting with red pine (*Pinus resinosa* Ait.). Key pedogenic processes were identified and rates and magnitude of soil recovery were quantified in terms of morphological, physical and chemical characteristics of soil profiles. Soil degradation generally involved reduced fertility with profile simplification (haploidization) on non-eroded fallowed fields, and topsoil loss by wind erosion (deflation) on more exposed eroded fields. After reforestation, soil restoration was characterized by cessation of erosion, accelerated horizon development and differentiation, reduced soil bulk density, and increased fertility and acidification of the soil. Chronofunctions revealed substantial recovery in soil organic C, total N, available P, and exchangeable K, Ca and Mg status within 75 yr of initial reforestation on non-deflated, fallowed sites. In contrast, estimated recovery of these parameters on severely deflated sites was delayed far beyond plantation maturity.

Key words: Oak Ridges Moraine, plantation (red pine), ecosystem restoration, deflation, soil degradation

McPherson, T. S. et Timmer, V. R. 2002. **Amélioration des sols dégradés dans les pinèdes rouges de la moraine d’Oak Ridges, en Ontario.** Can. J. Soil Sci. **82**: 375–388. Les auteurs ont étudié la dégradation du sol et sa restauration subséquente dans les chronoséquences « forêt ancienne, champ abandonné et peuplement de conifères jeune à mature » de la moraine d’Oak Ridges, relief à l’environnement fragile situé près de Toronto et que menace l’urbanisation. Les chronoséquences révèlent un déboisement avant la colonisation, l’exploitation agricole du sol par les pionniers, l’abandon des terres puis leur retour à l’état sauvage et/ou leur érosion par le vent dans les années 20 suivi par la stabilisation du sol après établissement d’un important peuplement de pins rouges (*Pinus resinosa* Ait.). Les auteurs ont identifié les principaux processus pédogénétiques puis calculé la rapidité et l’importance du rétablissement du sol d’après les propriétés morphologiques, physiques et chimiques de son profil. La dégradation du sol suppose généralement une baisse de fertilité associée à la simplification du profil (haploïdisation) dans les champs en jachère non érodés et à la perte de la couche de surface par l’érosion éolienne (déflation) dans les champs érodés les plus exposés. Après reboisement, la restauration se caractérise par un arrêt de l’érosion, la genèse et une différenciation plus rapides des horizons, une réduction de la masse volumique apparente ainsi qu’une plus grande fertilité et acidification du sol. Les fonctions chronologiques révèlent un renouvellement appréciable du carbone organique, du N total, du P disponible et des ions K, Ca et Mg échangeables dans les 75 années suivant le reboisement initial, pour les champs en jachères non érodés par le vent. Il faut cependant attendre longtemps après le moment où le peuplement parvient à maturité pour voir les mêmes paramètres se rétablir dans les champs très érodés.

Mots clés: Moraine d’Oak Ridges, peuplement (pin rouge), rétablissement d’un écosystème, déflation, dégradation du sol

Potential environmental impacts of recent, unprecedented urbanization are a source of growing apprehension regarding the integrity of the Oak Ridges Moraine (Fig. 1), a prominent landform complex of southern Ontario (Barnett et al. 1998). Proposed developments have raised concerns over the ecological stability of the Oak Ridges Moraine and possible effects on soil and water resources (Oak Ridges Moraine Technical Working Committee 1994; Howard et al. 1996; Sharpe et al. 1996). The concerns are founded on a long history of land use that has proven the Oak Ridges Moraine to be a fragile landform, largely due to the sensitivity of the predominant glacial outwash soils to disturbances (Kelly 1974; Hill 1976). Extensive deforestation

after early European settlement followed by cultivation and pasturing of the land resulted in widespread soil degradation during the late 1800s and early 1900s, leading to eventual abandonment of farmlands (Carman 1941; Chapman and Putnam 1984; Wood 1991).

This pattern of land use often resulted in loss of soil organic matter (Houghton et al. 1983; Schlesinger 1990) and concomitant decreases in nutrients, leaving areas of limited soil fertility (Rolfe and Boggess 1973; Inouye et al. 1987; Nowak et al. 1991; Quideau and Bockheim 1996). On many areas of the Oak Ridges Moraine, however, agriculture followed by abandonment also resulted in extensive wind erosion (i.e., deflation: removal of fine soil particles by

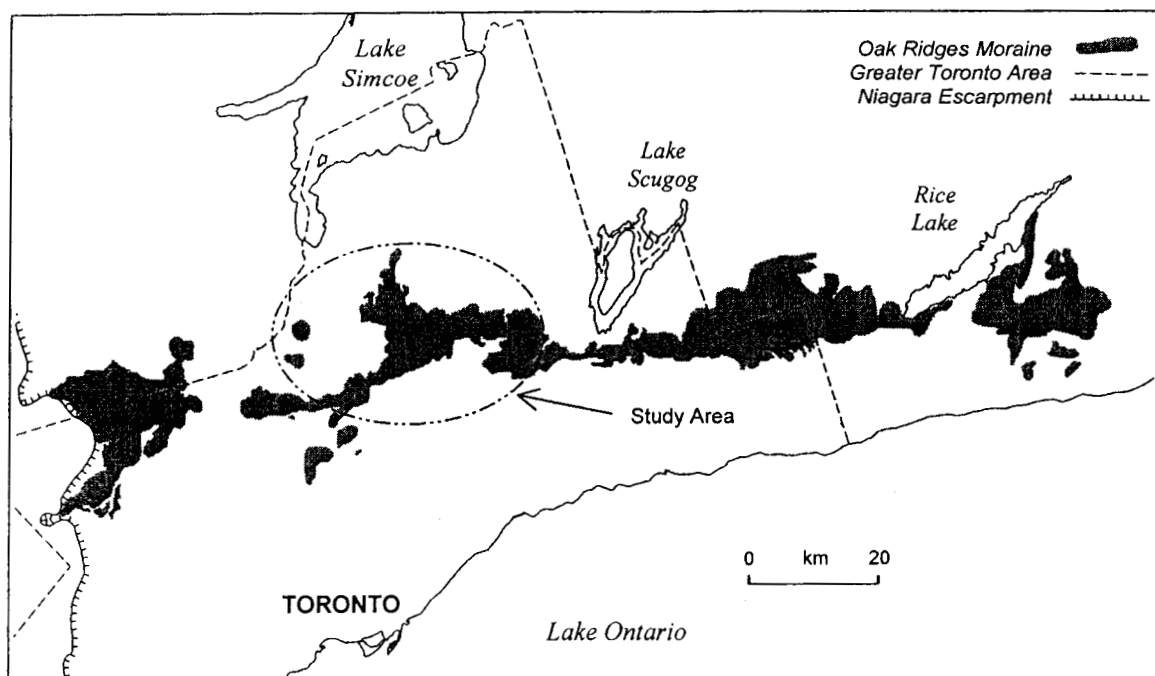


Fig. 1. Location of the Oak Ridges Moraine (centre is approximately 44°N latitude, 79°W longitude) and study area [adapted from Sharpe et al. (1994) and Dyke et al. (1997)].

wind), creating “wastelands” through topsoil removal and consequent exposure of calcareous parent material (Zavitz 1960; Hill 1976). These wastelands were later reforested to stabilize the drifting soils (Kuhlberg 1996). Native red pine (*Pinus resinosa* Ait.) seedlings were most commonly planted, because of the relative tolerance of this species to dry and nutrient poor conditions (Wilde 1966; Fisher and Stone 1968; Parker et al. 2001).

In addition to contributing to erosion control, reforestation and subsequent plantation development played an important role in ecosystem restoration on degraded, abandoned agricultural lands not conducive to natural regeneration (Rolfe and Boggess 1973). It is well known that plantation establishment facilitates natural forest regeneration by ameliorating microclimate and soil conditions, as demonstrated in tropical forest restoration (Fisher 1995; Lugo 1997; Parrotta et al. 1997; Bhojvaid and Timmer 1998). This effect occurred under red pine plantations on various soil conditions (Wilde 1961, 1964; Wilde and Iyer 1962; Nowak et al. 1991), but on severely degraded sites, such as extensively deflated soils, the process could be exceedingly slow. Reduced soil productivity may persist beyond plantation maturity (Farrish 1987), which could well be the case on sites of the Oak Ridges Moraine. Considering the environmental sensitivity of the region and the current development pressures on the land, it is important to know for planning purposes the resiliency and restoration potential of disturbed sites on this landform. The intention of this study was to retrospectively examine degradation and amelioration processes of severely disturbed soils in the area, and determine relative recovery rates after reforestation. The focus was on examining two disturbance regimes after deforestation: the

moderately degraded soils that were fallowed and not eroded after abandonment, and the severely degraded soils that were both fallowed and deflated after abandonment. Red pine was planted extensively on abandoned farmlands of the Oak Ridges Moraine since 1924, providing an array of differently aged plantations that can now be used to evaluate restoration processes of degraded soils employing a chronosequence approach.

METHODS AND MATERIALS

The Chronosequence Approach

Direct measurement of pedogenic changes in soils is often impractical and difficult to accomplish in a limited period; hence retrospective techniques can be practical alternatives. The use of chronosequences is recognized as an effective retrospective approach to study soil development (Stevens and Walker 1970) and elucidate multidirectional pathways of soil evolution (Huggett 1998). A chronosequence is defined as a suite of genetically related soils that evolved under similar conditions of climate, topography and vegetation, in which spatial differences between soils are translated to temporal differences (Huggett 1998).

The presence of degraded and recovering soils of different ages on the Oak Ridges Moraine offered an opportunity to examine disturbance impacts and the extent of soil recovery along a chronosequence spanning several stages of soil degradation and restoration. Conceptually, soil degradation started with tree removal in the natural “Forest” stage that represented relatively undisturbed pre-settlement conditions, progressing to the “Deforestation and Agriculture” stage reflecting periods of farming and soil exploitation that

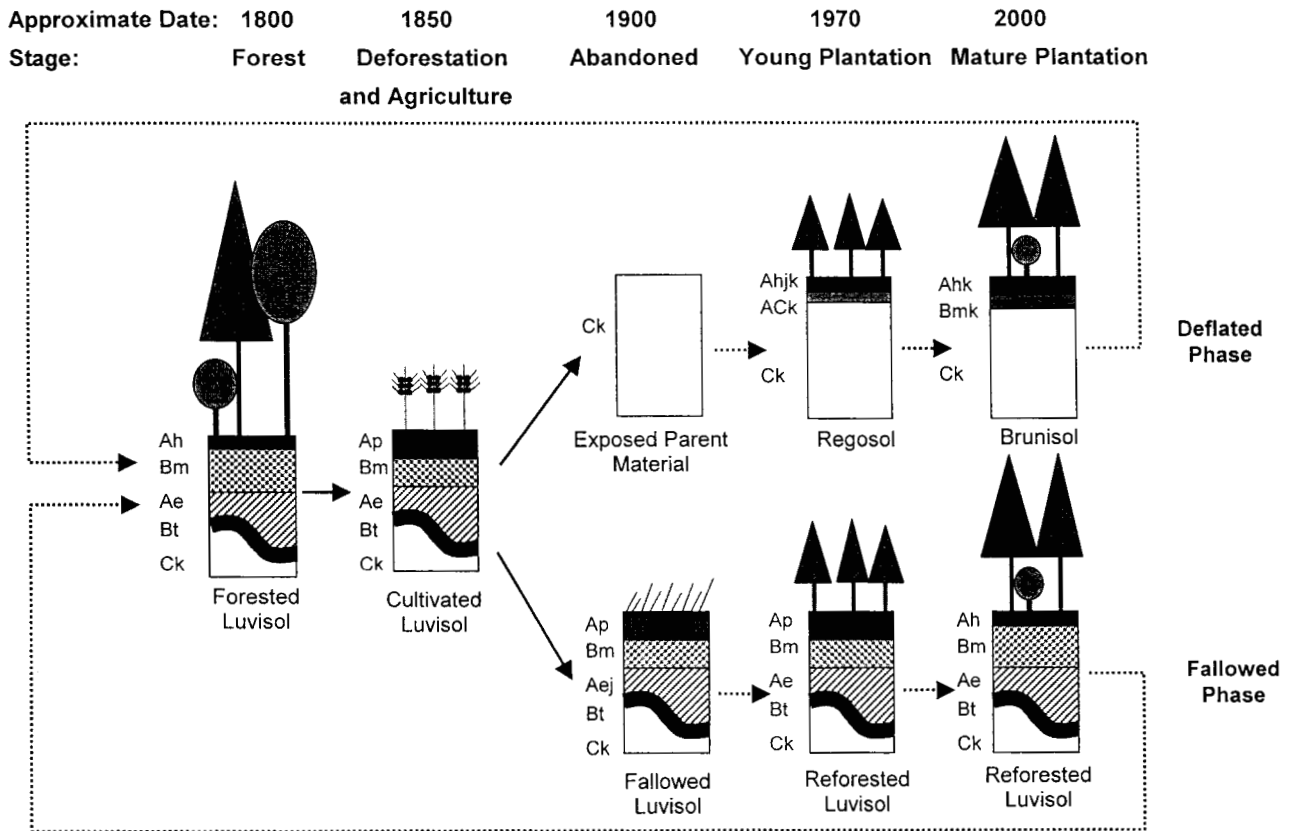


Fig. 2. Conceptual chronosequence of soil degradation and restoration on the Oak Ridges Moraine induced by pre-settlement deforestation, followed by exploitive pioneer agriculture, land abandonment and fallowing and/or deflation (solid arrows). Soil restoration (dotted arrows) was initiated by reforestation with red pine that will likely progress towards the Forest stage (Luvisol) assuming environmental constancy and the absence of further disturbance. Dominant coniferous and deciduous vegetation are illustrated by triangular and elliptical crowns; crops and grasses usually occurred on Ap horizons.

Table 1. Tree and shrub characteristics of each stage of the chronosequence (each row represents an average from three sites)

Stage/phase	Canopy ^a			Years since planting	Understorey		
	Density (stems ha ⁻¹)	DBH ^b (cm)	Height (m)		< 1 m (% cover)	1–2 m (stems ha ⁻¹)	> 2 m (stems ha ⁻¹)
<i>Forest</i>							
N/A	236 ± 112	51.3 ± 8.15	25.4 ± 3.30	N/A	50	200	600
<i>Abandoned</i>							
Deflated ^x	N/A	N/A	N/A	N/A	No vegetation > 90 % grasses, < 10% forbs		
Fallowed	N/A	N/A	N/A	N/A			
<i>Young plantation</i>							
Deflated	1211 ± 465	22.1 ± 5.36	16.6 ± 2.44	30–54	< 5	0	0
Fallowed	1256 ± 633	19.2 ± 4.45	15.7 ± 2.43	30–50	< 5	100	0
<i>Mature plantation</i>							
Deflated	433 ± 150	30.9 ± 3.17	22.9 ± 2.67	69–71	38	300	833
Fallowed	510 ± 88	28.5 ± 3.08	24.5 ± 2.15	61–75	27	333	733

^aOverstorey stand density, tree DBH, and tree height are recorded ± one standard deviation of the mean.

^bDBH is diameter at breast height (measured 1.3 m above the ground).

^xDeflated refers to soils that were eroded by wind (in this case topsoil and subsoil removal, and exposure of parent material).

led to an “Abandonment” stage when soils were fallowed (Fig. 2). Subsequent soil recovery under reforestation in “Young” and “Mature” stages of plantation development usually occurred on two types of post-abandoned sites: the

less-disturbed fallowed soils (Fig. 2, bottom phase), and the severely disturbed deflated soils (Fig. 2, top phase).

In this study, the existing natural deciduous forest with white pine (*Pinus strobus* L.) often in the upper canopy

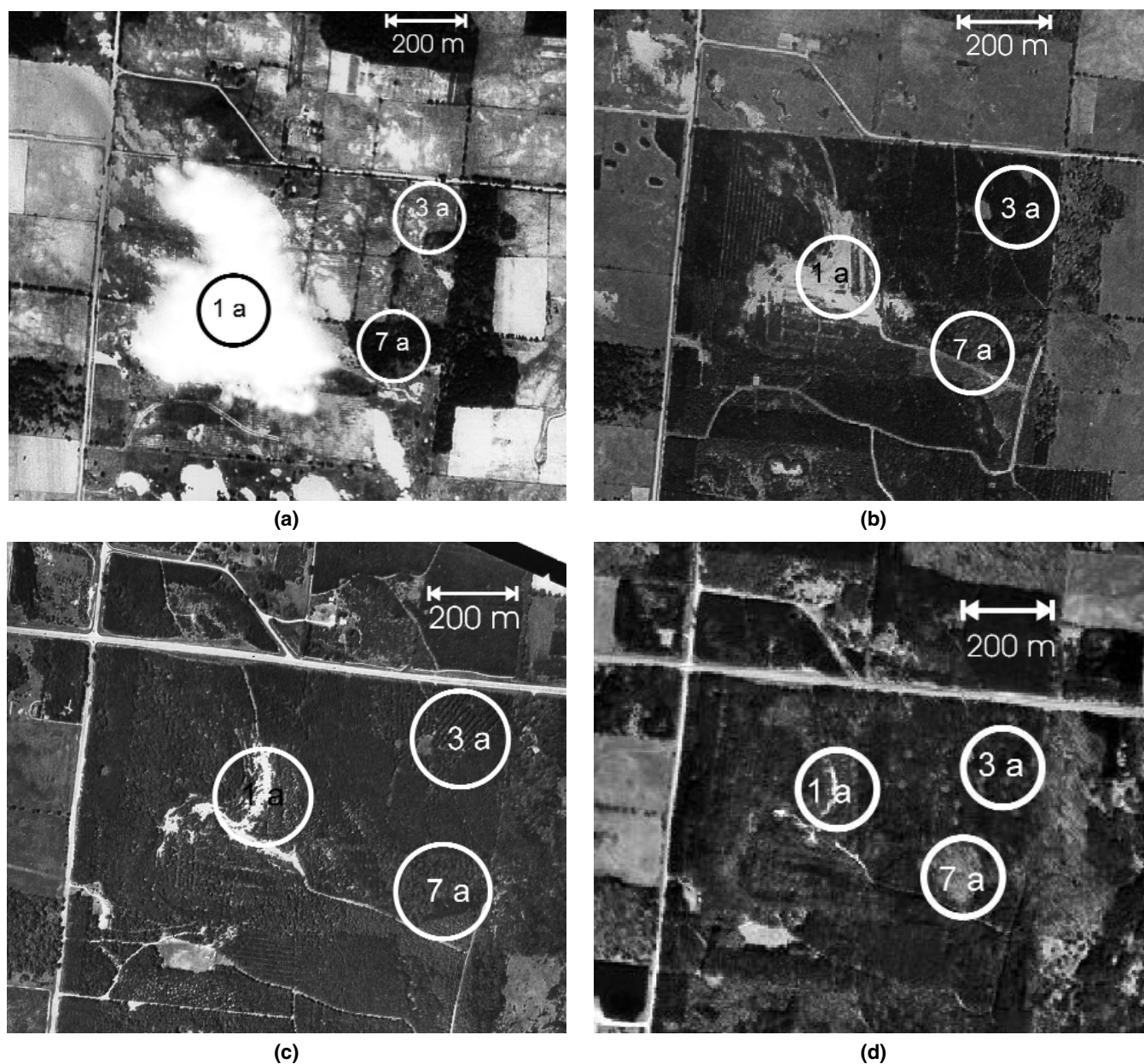


Fig. 3. Aerial photograph sequence of a portion of the Durham Regional Demonstration Forest: (a) 1927, prior to planting (National Aerial Photograph Library, Ottawa), (b) 1954, (Ministry of Natural Resources, Toronto) (c) 1978 (Ministry of Natural Resources, Toronto), (d) 1997, photographed in spring prior to leaf development (B. Hatton Photography, Buttonville). The circled numbers represent one partial replication of the deflated phase of the chronosequence: (7a) Forest stage, represented by remnant forest; (3a) Mature Plantation stage, represented by a deflated area reforested with red pine; and (1a) Abandoned stage, represented by a deflated area that remains unvegetated, with exposed parent material.

exemplified “old growth” characteristics (Oliver and Larson 1996), and represented the initial Forest stage of the chronosequence. This stage was assumed to resemble pre-settlement conditions of forests and soils within the central portion of the moraine complex. Abandoned soils caused by deforestation, unsustainable farming and subsequent fallowing were represented by current derelict land essentially free of vegetation, or dominated by grasses. The deflated soils (including those later reforested) were identified from

archival aerial photographs showing distinct blow-outs with calcareous parent material exposed. The fallowed sites were selected on the basis of no sign of past erosion. The recovery stages were evaluated under consistent site conditions and tree species (planted red pine): Young Plantations were <55 yr old and generally lacked understorey vegetation, and Mature Plantations were >60 yr old and supported vigorous deciduous understorey vegetation (Table 1). These structural characteristics matched the “stem exclusion” and “under-

Table 2. Soil profile characteristics of each stage of the chronosequence (each row represents an average of three sites)

Stage/phase/ soil classification	Horizon and depth (cm)	Fine earth fraction (g kg ⁻¹)			Structure	Consistency	Munsell colour (moist)
		Sand	Silt	Clay			
<i>Forest</i>							
Brunisolic	Ah (0-9)	796	145	59	Fine crumb	Very friable	10YR 2/1
Gray Brown	Bm (9-35)	918	39	43	Very weak platy	Very friable	10YR 4/4 (3/4)
Luvisol	Ae (35-66)	848	124	28	Single grain	Loose	10YR 5/3 (5/4)
	Bt (66-71)	835	77	88	Weak subangular blocky	Friable	10YR 4/4 (4/3)
	Ck (71-130+)	935	49	16	Single grain	Loose	10YR 5/2
<i>Abandoned, deflated</i>							
Orthic	Ck (0-130+)	941	36	23	Single grain	Loose	10YR 5/2
Regosol							
<i>Abandoned, fallowed</i>							
Brunisolic	Ap (0-21)	735	226	39	Fine crumb	Very friable	10YR 3/2
Gray Brown	Bm (21-32)	734	225	40	Very weak platy	Very friable	10YR 4/3 (3/3)
Luvisol	Aej (32-58)	764	200	37	Single grain	Loose	10YR 4/3 (4/4)
	Bt (58-68)	715	173	109	Weak subangular blocky	Friable	10YR 4/4 (4/3)
	Ck (68-130+)	852	127	21	Single grain	Loose	10YR 5/2
<i>Young plantation, deflated</i>							
Orthic	Ahk (0-5)	880	92	28	Fine crumb/weak platy	Very friable	10YR 4/1 (3/1)
Regosol	ACk (5-15)	928	47	25	Very weak platy	Very friable	10YR 4/2 (5/2)
	Ck (15-130+)	898	79	23	Single grain	Loose	10YR 5/2
<i>Young plantation, fallowed</i>							
Brunisolic	Ap (0-21)	887	75	37	Fine crumb/weak platy	Very friable	10YR 3/1 (2/1)
Gray Brown	Bm (21-34)	906	60	35	Very weak platy	Very friable	10YR 4/4 (4/3)
Luvisol	Aej (34-61)	913	53	33	Single grain	Loose	10YR 5/4 (4/4)
	Bt (61-71)	825	69	106	Weak subangular blocky	Friable	10YR 3/3 (3/4)
	Ck (71-130+)	934	39	27	Single grain	Loose	10YR 5/2
<i>Mature plantation, deflated</i>							
Orthic Eutric	Ahk (0-6)	913	52	35	Fine crumb	Very friable	10YR 3/1 (2/1)
Brunisol	Bmk (6-17)	946	35	19	Very weak platy	Very friable	10YR 4/3
	Ck (17-130+)	929	53	19	Single grain	Loose	10YR 5/2
<i>Mature plantation, fallowed</i>							
Brunisolic	Ah (0-14)	879	72	49	Fine crumb	Very friable	10YR 2/1 (3/1)
Gray Brown	Bm (14-37)	927	35	38	Very weak platy	Very friable	10YR 3/4 (4/4)
Luvisol	Ae (37-68)	936	39	25	Single grain	Loose	10YR 5/4
	Bt (68-73)	905	34	61	Weak subangular blocky	Friable	10YR 4/3
	Ck (73-130+)	959	26	15	Single grain	Loose	10YR 5/2

storey reinitiation" stages of stand development described by Oliver and Larson (1996).

An example of site selection in the Durham Regional Forest is shown with archival aerial photographs taken in 1927, 1954, 1978 and 1997 (Fig. 3). Site 7a in a natural forest represented the Forest stage. A plantation established on deflated soils (white-coloured blow-outs at site 3a and later diagonally thinned [photograph (c)] represented the Mature Plantation stage with abundant understorey vegetation. A persistently deflated area (site 1a) free of vegetation with exposed parent material represented the deflated Abandoned stage. The deflated Young Plantation stage, the fallowed phase, and additional replications were located in the surrounding area, beyond the area covered in these photographs.

Description of Location

A total of 21 sites were selected on the Oak Ridges Moraine (Fig. 1) within a 20-km radius to minimize climatic differ-

ences. Each chronosequence was replicated three times, consisting of seven sites that included a Forest stage for reference, and an Abandonment, a Young Plantation and a Mature Plantation stage, each established on two disturbance phases – fallowed or deflated soils. The sites were situated on upland positions with <10% slope. Mean daily temperature for the year ranges from 5.5 to 7.8°C with mean annual precipitation of 762 to 813 mm (Brown et al. 1980). To ensure parent material homogeneity within and between sites, sample plots were located on similar calcareous, sandy outwash parent material of the Pontypool and Brighton soil series (Hoffman and Richards 1990; Olding et al. 1990). At each site a 100-m² circular plot was set up to determine size-class, density and species composition of understorey trees and shrubs as well as density and diameter at breast height (DBH) of overstorey trees. Tree height measurements were taken of at least nine trees per site. Plantation ages were determined from planting dates noted in management

Table 3. Humus characteristics of forest and plantation sites (each row is a summary of three sites)

Stage/phase	Humus classification	Horizon	Horizon thickness (cm)		Deciduous litter (%)	Primary decomposition agents
			Min.	Max.		
<i>Forest</i>						
N/A	Verimull and mullmoder	L (Fa)	1.0 0	5.0 0.5	> 75	Fauna > fungi
<i>Young plantation</i>						
Deflated	Hemimor, mormoder and mullmoder	L Fa/Fm	2.0 0.5	4.0 2.0	0	Fungi > fauna
Fallowed	Mullmoder	L Fz (Hh)	2.0 0.5 0	3.0 2.5 0.5	< 5	Fauna > fungi
<i>Mature plantation</i>						
Deflated	Mullmoder, mormoder and verimull	L Fa/Fz (Hh)	2.0 0.5 0	3.0 2.5 0.5	10–30	Fauna > fungi
Fallowed	Mullmoder and verimull	L Fa (Hh)	2.0 0.5 0	3.0 2.0 0.5	10–30	Fauna > fungi

records, and were confirmed by ring counts from tree increment cores.

Soil Sampling and Classification

A representative soil pit was excavated at each site to characterize soil horizons and humus layers. A depth of 1.3 m was required to reach parent material of the deeper, undisturbed and fallowed soils and was used for consistency at all sites. Composite soil samples were taken from the centre of each horizon, from all four sides of the soil pit, and sample depth recorded. At most, five horizons were identified, and where fewer than five horizons were present, samples were taken from comparable depth intervals. Each solum was classified according to the Canadian System of Soil Classification (Soil Classification Working Group 1998) and humus forms were classified according to the more detailed system outlined by Green et al. (1993). Moist colours were determined using a Munsell chart and carbonate presence was tested by reaction with 10% HCl.

Laboratory Procedures

Soil samples were air dried and sieved with a 2-mm screen; oven-dry mass and residual water content were determined by drying at 105°C. Particle size distributions were determined by the sedimentation-hydrometer method (Day 1965). Bulk density was determined by extracting samples with a copper core, as described by Campbell and Henshall (1991). Soil pH was measured by glass electrode in a saturated soil water paste. Oxidizable (organic) C was determined by the Walkley-Black method (Nelson and Sommers 1996), total N by Kjeldahl digestion and auto-analyzer (Schuman et al. 1973), available P by Bray's NH_4F extraction (Bray and Kurtz 1945), and exchangeable cations (K, Ca and Mg) by neutral 1 M ammonium acetate extraction and atomic absorption spectroscopy (Thomas 1982).

Experimental Design and Statistics

A completely randomized $7 \times 5 \times 3$ factorial design was employed to evaluate data derived from seven chronosequence stages at five profile depths replicated three times. A two-way analysis of variance, with $\alpha = 0.05$, was carried out using MS Excel® software to test for differences between means of chronosequence stages, sample depths, and combined stage and depth responses for each soil parameter. Differences between chronosequence stages were further examined with chronofunctions commonly adopted to quantify soil development and recovery rates of disturbed soils (Bockheim 1980; Schaetzl et al. 1994). Chronofunctions were formulated for deflated and fallowed phases of the soils by regressing soil parameters with time since reforestation. Soil parameters were expressed as mass per hectare: the product of bulk density multiplied by corresponding soil parameters' concentration for each sample depth as described by Schlesinger (1990). Because this method is not applicable for pH and bulk density, only the surface values were used in chronofunction of these parameters.

RESULTS AND DISCUSSION

Morphological characteristics of the mineral soil horizons and humus layers are summarized in Tables 2 and 3, respectively; soil fertility parameters are presented in Figs. 4 and 5. Significant differences ($P < 0.001$, $\alpha = 0.05$) occurred between means of all soil fertility parameters for both chronosequence stages and sample depths (Figs. 4 and 5). The combined interactions of the analysis of variance were also significant for all parameters ($P < 0.001$, $\alpha = 0.05$), except for exchangeable Mg ($P = 0.08$) and exchangeable K ($P = 0.69$) status, suggesting these two parameters were less sensitive as indicators of the chronosequence. Parent material (Ck) characteristics of the chronosequence soils were all similar, supporting assumptions that the sample soils from

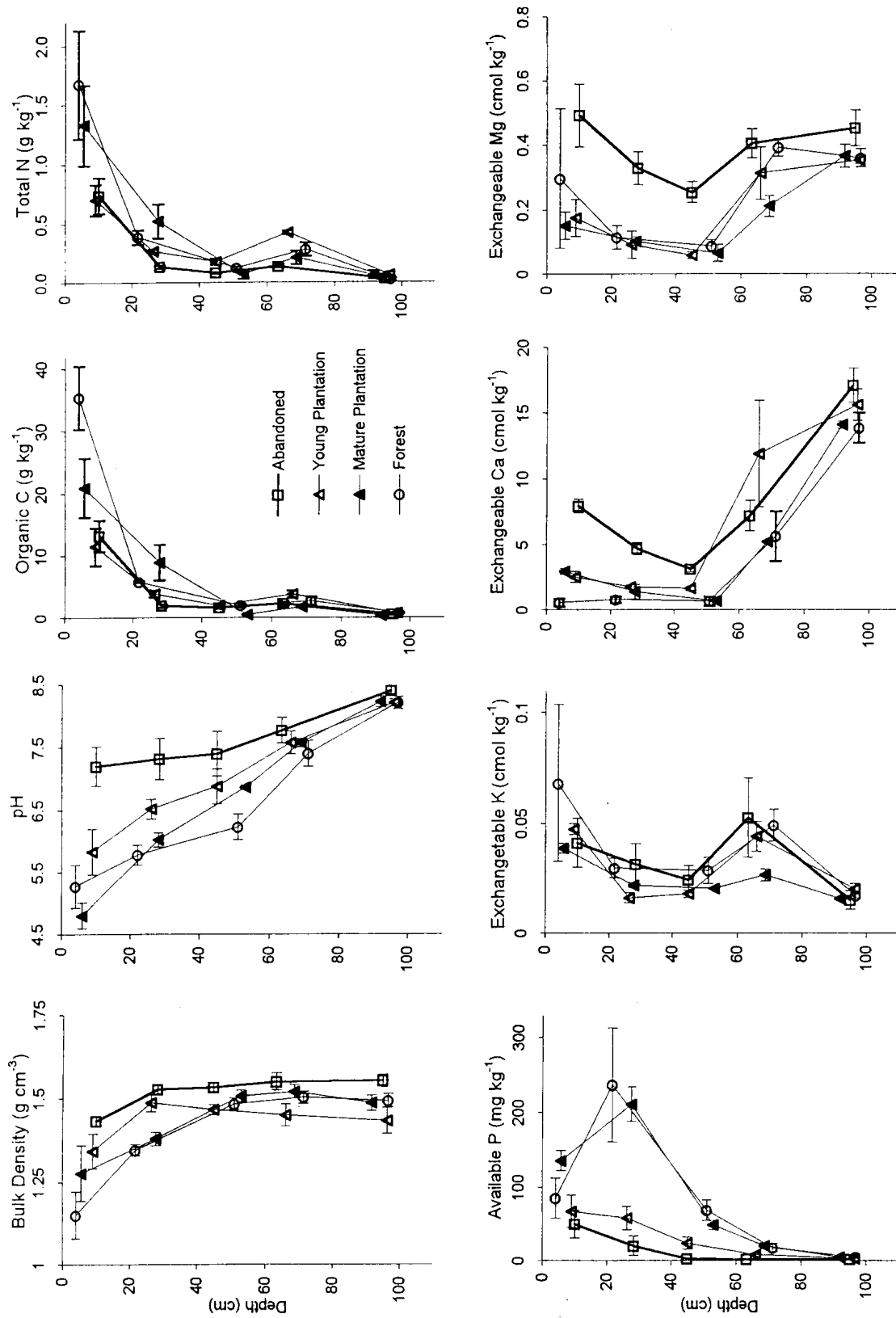


Fig. 4. Fallowed sites: profile distributions of soil parameters illustrating effects of degradation (Forest to Abandoned) and recovery (Abandoned to Young Plantation to Mature Plantation). Horizontal bars represent the standard error of each mean.

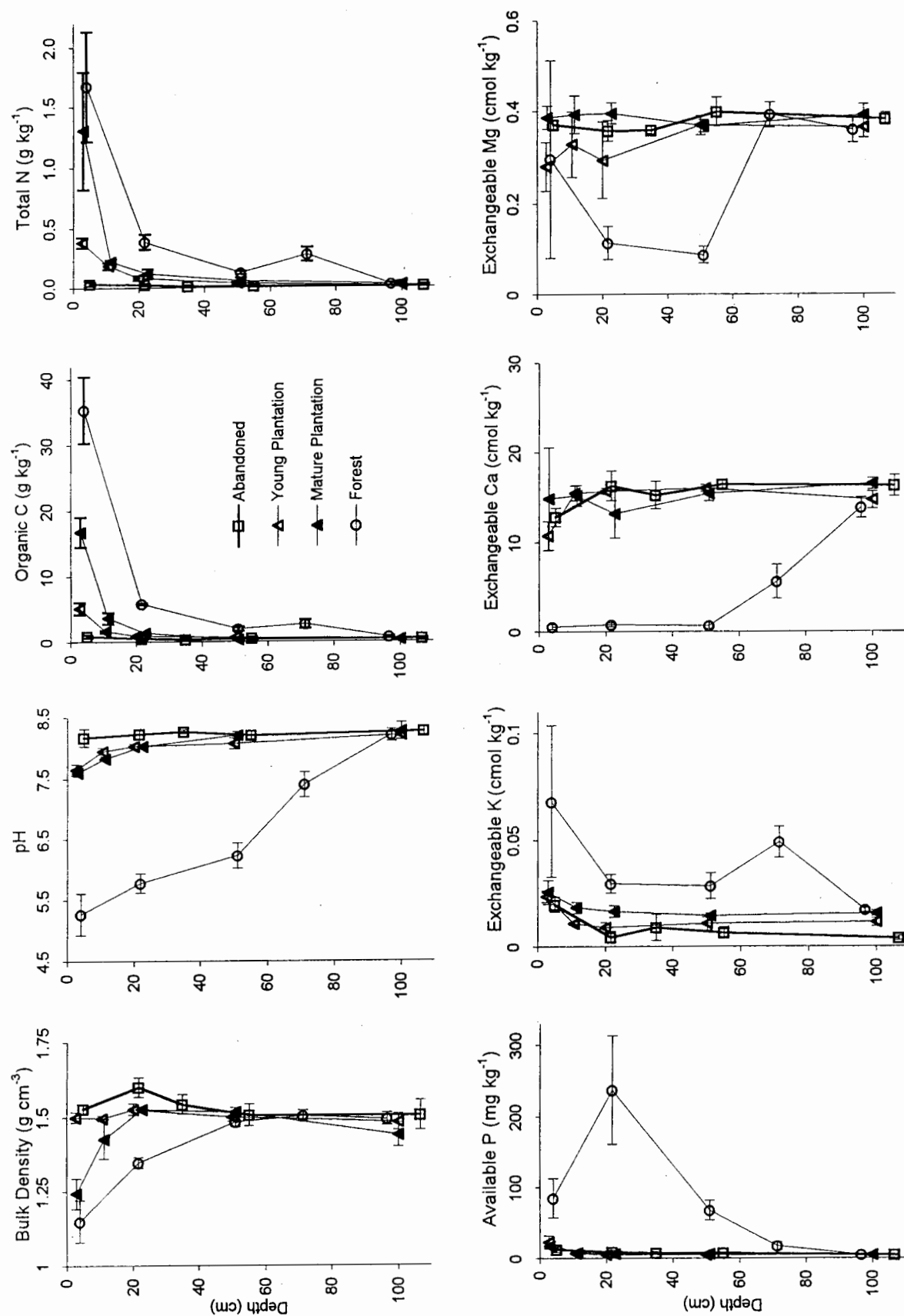


Fig. 5. Deflated sites: profile distributions of soil parameters illustrating effects of degradation (Forest to Abandoned) and recovery (Abandoned to Young Plantation to Mature Plantation). Horizontal bars represent the standard error of each mean.

each phase were likely derived from the same parent material. Apparently, land use practices impacted topsoils more strongly than subsoils resulting in larger parameter differences in upper horizons compared to lower horizons along the full chronosequence (Figs. 4 and 5). Associated standard errors were also larger for surface horizons compared to subsoil horizons.

Natural Forest Stage

Because of lack of prior disturbance it was assumed that the initial Forest stage of the chronosequence reflected soil conditions before deforestation and pioneer agriculture. Accordingly, this stage served as the reference base for impact assessment of subsequent land use practices (Fig. 2). The soils of the Forest stage were classified as Brunisolic Gray Brown Luvisols, typical for the study area under forested conditions (Stobbe 1952; Hoffman and Richards 1990) [taxonomic terms and horizon designations have been updated according to Clayton et al. (1977) and the Soil Classification Working Group (1998)]. The horizon sequence of Ah, Bm, Ae, and a clay-enriched Bt overlying highly calcareous parent material (Ck) (Table 2) was visually distinct due to pronounced colour, texture and structure differences. The blackish Ah horizon, capped by thin verimull and mullmoder humus forms, was crumbly in structure and finer in texture than the subsoil horizons (Tables 2 and 3). This horizon was relatively low in bulk density (1.15 g cm^{-3}) and relatively high in organic C content through litter incorporation and plant root activity (Fisher and Binkley 2000). Bulk density, however, increased with depth to 1.49 g cm^{-3} in the parent material (Figs. 4 and 5). The transition was diffuse between the Bm and Ae horizons, but abrupt between the Ae horizon and the thin argillic Bt horizon. The eluvial-illuvial sequence presumably formed as a result of leaching and flocculation of clay above the calcareous parent material following upper solum decalcification (Runge 1973; Schaetzl 1996). Consequently, the boundary between the solum and the carbonate-rich Ck was abrupt and well defined, although wavy (50 to 130 cm) with an average depth of 71 cm (Table 2).

Soil chemical properties reflected the morphological differences between topsoil, subsoil and parent material (Figs. 4 and 5). Inputs of organic acids, organic matter and nutrients to the topsoil presumably lowered soil reaction (pH) from 8.2 in the parent material to 5.3 at the surface, but elevated topsoil organic C (35.4 g kg^{-1}), total N (1.6 g kg^{-1}), and exchangeable K ($0.068 \text{ cmol kg}^{-1}$) and Mg ($0.297 \text{ cmol kg}^{-1}$) status. Exchangeable Ca levels, however, were similar between the topsoil ($0.56 \text{ cmol kg}^{-1}$) and the underlying Bm horizon ($0.77 \text{ cmol kg}^{-1}$), possibly because of rapid leaching in the acidic environment of the surface horizons (Lichter 1998) and/or higher retention than return of Ca with some tree species (Alban 1982). Available P was highest in the Bm horizon (237 mg kg^{-1}), presumably due to greater P solubility under more neutral soil pH compared to the acidic Ah horizon (Hausenbuiller 1985). Organic C, total N and exchangeable K, Ca and Mg levels were higher in the Bt horizon than the overlying mineral horizons reflecting illuviation processes. The relatively high cation exchange

capacity of the Bt was associated with clay enrichment (Hausenbuiller 1985). Elevated levels of exchangeable Ca and Mg in the calcareous parent materials may also reflect an artefact of dissolved carbonates from the soil during ammonium acetate extractions (Hendershot et al. 1993).

Abandoned Stage

Examples of past agricultural soils following the deforestation period (Fig. 2) were no longer available for study in the area because of adoption of modern farming practices. However, expected impacts on topsoil conditions would be conversion of Ah to Ap horizons associated with organic matter and nutrient loss from cropping (Houghton et al. 1983; Quideau and Bockheim 1996), and increased bulk densities caused by the loss of tree root activity (Rolfe and Boggess 1973; Fisher 1995) and compaction by surface activity. Due to the inherently low agricultural capability of the Pontypool series (Hoffman and Richards 1990; Olding et al. 1990), agricultural activity on any given parcel of land was likely brief. Hence farm fields were often abandoned in fallowed or eroded conditions (Fig. 2) as reported by Carman (1941), Richards and Morwick (1942) and Howard et al. (1996).

Fallowed Phase

The abandoned fallowed fields usually reverted to grassland vegetation (Fig. 2). The soils likely retained surface Ap horizons, hence were classified as cultivated Brunisolic Gray Brown Luvisols (Table 2). However, profile differences between the Forest and Abandoned stages reflect a dominance of regressive pedogenesis during the reversion period (Johnson and Watson-Stegner 1987). Topsoil disruption by cultivation and melanization of the Ae horizon (Johnson and Watson-Stegner 1987) presumably led to blending of the upper solum and less colour distinction between Bm, Ae_j and Bt horizons. Although the Bt horizon was clearly differentiated by clay enrichment (Table 2), the overlying Bm and Ae_j horizons were less pronounced visually. Weaker horizon differentiation was ascribed to regression towards more simplified profiles, or haploidization (Johnson and Watson-Stegner 1987; Johnson et al. 1987) likely due to reduced eluviation in the profile.

Topsoil bulk density increased substantially from 1.15 g cm^{-3} under the natural forest condition to 1.43 g cm^{-3} in fallowed fields (Fig. 4). Compaction by machinery and livestock, loss of tree roots, reductions in faunal activity and accelerated organic matter decomposition probably contributed to the higher bulk density (Rolfe and Boggess 1973; Fisher 1995). The Ap horizon was also lighter in colour (Table 2) usually associated with organic C loss [from 35.4 to 13.3 g kg^{-1} (Fig. 4)]. Total N in topsoil also declined from 1.7 g kg^{-1} to 0.7 g kg^{-1} . The largest drop in P availability (237 mg kg^{-1} to 20 mg kg^{-1}) occurred in the Bm horizon where levels were naturally highest. Soil exchangeable K levels, on the other hand, remained relatively stable during the chronosequence. During old-field succession, Odum et al. (1984) noted that exchangeable K was stable on sandy sites following abandonment, while available P status decreased initially, but stabilized later.

The large (14-fold) increase in exchangeable Ca accompanied by a more moderate rise in exchangeable Mg was attributed to a regressive pedogenesis associated with accelerated base-cycling often occurring in forest-to-grassland conversions (Johnson and Watson-Stegner 1987). Higher base-cycling was also reflected by increased topsoil pH (by two units) due to neutralizing effects of added bases under fallow vegetation (Hausenbuiller 1985; Johnson and Watson-Stegner 1987).

Although weaker horizon differentiation and higher base-cycling noted in the upper solum may reflect regressive pedogenesis (Johnson and Watson-Stegner 1987), fallowing caused relatively little soil degradation beyond the Agriculture stage. Apparently grass and forb invasion soon after abandonment contributes to improved soil fertility through organic matter inputs (Paul and Collins 1998) and nutrient build-up in the mineral soil (Inouye et al. 1987; Gleeson and Tilman 1990). Thus rapid re-vegetation of these fallowed fields may have played an important role in the maintenance or partial recovery of soil fertility, as well as stabilizing topsoils and deterring erosion.

Deflated Phase

Without forest cover or protective vegetation from cropping or pasturing, exposed soils unsheltered from prevailing winds were subjected to deflation after abandonment (Fig. 2). Deflation, a regressive pedogenic process (Johnson and Watson-Stegner 1987), is a major contributor to soil degradation because organic matter and nutrient loss coincide with removal of topsoil and subsoil (Lyles 1975; Larney et al. 1998). Hence, exposed calcareous parent material prevalent on many parts of the moraine was considered essentially the result of unabated deflation that led to truncation of the former soil profile (Hill 1976). As Orthic Regosols, these profiles were very weakly developed, lacking distinct A and B horizons, as well as texture and structure changes with depth (Table 2). Bulk density and most chemical properties were relatively uniform in the profile, except for a slight increase in exchangeable K at the surface that may reflect leached K from wind-blown leaf fall from nearby forests (Fig. 5).

Plantation Stage

Fallowed Phase

Planting of red pine seedlings on abandoned soils initiated development of protective cover, litterfall addition and humus layer formation that reduced erosion and started the first major stage of site recovery (Fig. 2). Change in profile morphology involved increased definition and differentiation of existing soil horizons. The Ap horizon, formed by past agricultural activity, evolved to a thinner and darker coloured Ah horizon (Table 2) because of surficial organic matter inputs. Soil organic C increased from 13.3 g kg⁻¹ at the Abandoned stage to 20.9 g kg⁻¹ in Mature Plantations (Fig. 4). A similar Ap–Ah horizon conversion was also noted by Billings (1938) after 30 yr of pine establishment, exemplifying the relatively rapid change that may occur in surface horizons under plantation development.

Topsoil organic matter of fallowed fields built up much faster in the Mature Plantation compared with the Young Plantation stage because of greater deciduous leaf litter (Millar 1974; Williams and Gray 1974) as plantations matured (Table 1). The forest floor developed more pronounced verimull characteristics with time (Table 3) coinciding with increased invasion of trees and shrubs in the understorey (Table 1). Higher organic matter incorporation presumably lowered topsoil bulk density from 1.43 g cm⁻³ at the Abandoned stage to 1.34 g cm⁻³ under Young Plantations and 1.28 g cm⁻³ under Mature Plantations, a phenomenon also noted by Fisher (1995).

Increased humus incorporation also contributed to soil nutrient replenishment (Runge 1973; Paul and Collins 1998). Topsoil total N almost doubled since the Abandoned stage, reaching 1.3 g kg⁻¹ at the Mature Plantation stage. Available P increased from 20 mg kg⁻¹ in the Bm horizon to 211 mg kg⁻¹, which closely approximated the value for the natural forests (237 mg kg⁻¹) (Fig. 4). Replenishment may primarily occur as a result of “nutrient pumping” since red pine develops extensive root systems (Fayle 1975), acquiring nutrients from large volumes of subsoil that are later re-deposited on the forest floor, thus concentrating nutrients in the surface horizons (Fisher 1990). Increased P availability in the Bm horizon may also reflect a rhizosphere effect, possibly through enzyme activity or changes in soil acidity at the root-soil interface, leading to Fe and Al chelation that promotes P solubility (Fisher and Stone 1969; Fisher 1990, 1995). The status and distribution of base cations in the soil profile were also similar in corresponding horizons of the Mature Plantation and Forest stages (Fig. 4). Exchangeable K remained essentially unchanged during the chronosequence, while reductions in exchangeable Ca and Mg levels noted in the upper half of the solum (to about one-third the status for the Abandoned stage) suggest a return to lower base-cycling in the plantation system.

For the full chronosequence period, the Bt horizons appeared relatively resilient to disturbance since levels of organic C, total N, and exchangeable K, Ca and Mg did not change appreciably (Fig. 4). The Bt horizons were formed deep in the profile, thus were relatively unaffected by pedoturbations from ploughing and possibly faunal activity. Apparently, agricultural activity and subsequent abandonment had little effect below 60 cm depth, while extensive restoration of physical and chemical properties occurred in the upper solum during plantation development and maturation.

Deflated Phase

In contrast to the fallowed soils, horizon development in deflated soils was much less pronounced (Table 2, Fig. 2). Since pedogenesis started on newly exposed parent material when deflation ceased, changes in morphological properties were restricted to the upper 15 to 20 cm of mineral soil. Apparently, the trees stabilized the soils, providing protective cover and also contributed to the formation of humus layers (Table 3). In all likelihood, Orthic Regosols developed initially, forming distinct Ahk and ACk horizons at the Young Plantation stage (Fig. 2). Later, these soils matured to form Orthic Eutric Brunisols at the Mature Plantation stage characterized by Bmk horizons to 20 cm in depth (Table 2).

With time, bulk density of surface horizons dropped from 1.53 g cm^{-3} at the Abandoned stage to 1.50 g cm^{-3} under Young Plantations and 1.24 g cm^{-3} found under Mature Plantations (Fig. 5). This drop was attributed mainly to a build-up and incorporation of organic matter in the soil from decomposed litter and woody debris, as well as response to greater root growth and faunal activity as the plantations developed (Rolfe and Boggess 1973, Fisher 1995). Currently, surface bulk density in the Mature Plantation stage is only 0.09 g cm^{-3} more than that in the Forest stage. The most pronounced reduction in bulk density occurred 50 to 75 yr after the initial planting, which probably reflects increased root activity and organic matter incorporation from more readily decomposed broad-leaved litter following ingress of deciduous species under the plantation canopy (Millar 1974; Williams and Gray 1974). This phenomenon was also evident in the surface soils of the fallowed phase (Table 1).

Changes in soil fertility were confined to the newly formed surface horizons (Fig. 5), and were associated with organic matter addition and "nutrient pumping" (Fisher 1990). Organic C levels increased from 0.9 to 5.2 g kg^{-1} following establishment of Young Plantations, and reached 16.8 g kg^{-1} at the Mature Plantation stage. Similar accumulation patterns were evident in total N status of the profile. Exchangeable K levels exhibited comparatively smaller increases during this period (0.02 – $0.03 \text{ cmol kg}^{-1}$). However, available P status did not change appreciably, presumably because of formation of less soluble calcium phosphates at high soil pH (>7.5) (Smeck and Runge 1971; Smeck 1973; Hausenbuiller 1987). As expected, soil exchangeable Ca and Mg did not change significantly during plantation development, probably reflecting the influence of carbonate-rich parent materials. The dissolution of carbonates during the ammonium acetate extraction of samples may also have contributed to high readings of these cations (Hendershot et al. 1993).

In summary, soil ameliorative effects under plantation development were clearly evident on deflated sites, although changes were restricted to the upper 15 to 20 cm of the mineral soil resulting in development of weakly weathered, calcareous Orthic Eutric Brunisols by plantation maturity. These soils precede development of deeper, more weathered Eluviated Eutric Brunisols and Eluviated Dystric Brunisols that will later form clay enriched Btj horizons (VandenBygaart and Protz 1995). Eventually, it is expected that characteristic Luvisols will evolve exhibiting distinct eluvial-illuvial horizon sequences (Schaeztl 1996) similar to the original soil condition of natural forests (Fig. 2).

Chronofunctions

Linear chronofunctions were formulated to determine recovery rates of soil parameters for the deflated and fallowed phases starting after abandonment at time of reforestation (Fig. 6). Slope or regression coefficient comparisons reveal that recovery rates were higher on fallowed sites than deflated sites for soil pH, available P and exchangeable Ca and Mg. Soil organic C, total N and exchangeable K contents during the fallowed phase were

comparable to amounts present in corresponding natural forests. Thus, asymptotic levels representing a dynamic equilibrium may have been attained prior to, or shortly after reforestation.

Except for surface bulk density, soil parameters from deflated sites deviated widely from the Forest stage range delineated by dashed lines in Fig. 6. Thus, after almost 75 yr of plantation development soil organic C, total N and available P were about one-third, one-half and one-sixth of the lower range of the Forest stage, respectively. However, strong trends ($P < 0.05$) of recovery were present for the other parameters, except for exchangeable Ca, Mg and available P status. The small change in P availability was possibly the result of high pH conditions that reduce solubility in excess of 7.5 (Smeck 1973), as mentioned previously.

CONCLUSIONS

Soil degradation and amelioration were studied by examining changes in morphological, physical and chemical properties of soils along a chronosequence of old-growth forest, abandoned fields, and young and mature plantations of red pine on the Oak Ridges Moraine. On the sandy outwash soils of the study area, degradation involved the following major regressive pedogenic processes:

- 1) profile simplification, i.e., haploidization, resulting in less distinct horizon sequences, commensurate with organic matter and nutrient losses, increased surface soil bulk density, and higher pH and base-cycling on fallowed sites and;
- 2) localized deflation of unprotected areas, resulting in truncated soil profiles associated with increased bulk density, higher soil pH and nutrient removals.

After reforestation with red pine planting stock, progressive pedogenesis took place reversing soil degradation and promoting restoration. Soil restoration induced by tree planting was associated with the following major ameliorative processes:

- 1) cessation of erosion due to development of tree and litter cover;
- 2) soil horizon redefinition on non-eroded, fallowed sites, and horizon development (horizonation) on deflated sites;
- 3) bulk density reductions due to soil organic matter accretion and increased root and faunal activity and;
- 4) increased fertility and acidification (reduction of pH towards values of undisturbed forest soils) of the topsoil associated with nutrient pumping and higher deposition and decomposition of organic material.

Following land abandonment, chronofunctions of soil properties indicated that reforestation led to substantial recovery of soil fertility during plantation development on fallowed soils. As such, soil remediation close to corresponding natural forest conditions was achieved within 75 yr of initial reforestation. In contrast, low resilience was evident on the deflated, calcareous sites that delayed soil restoration well beyond the period of plantation maturation. The susceptibility of deflated sites to severe degradation and the limited potential for recovery soon after reforestation exemplifies the environmental sensitivity of forested soils

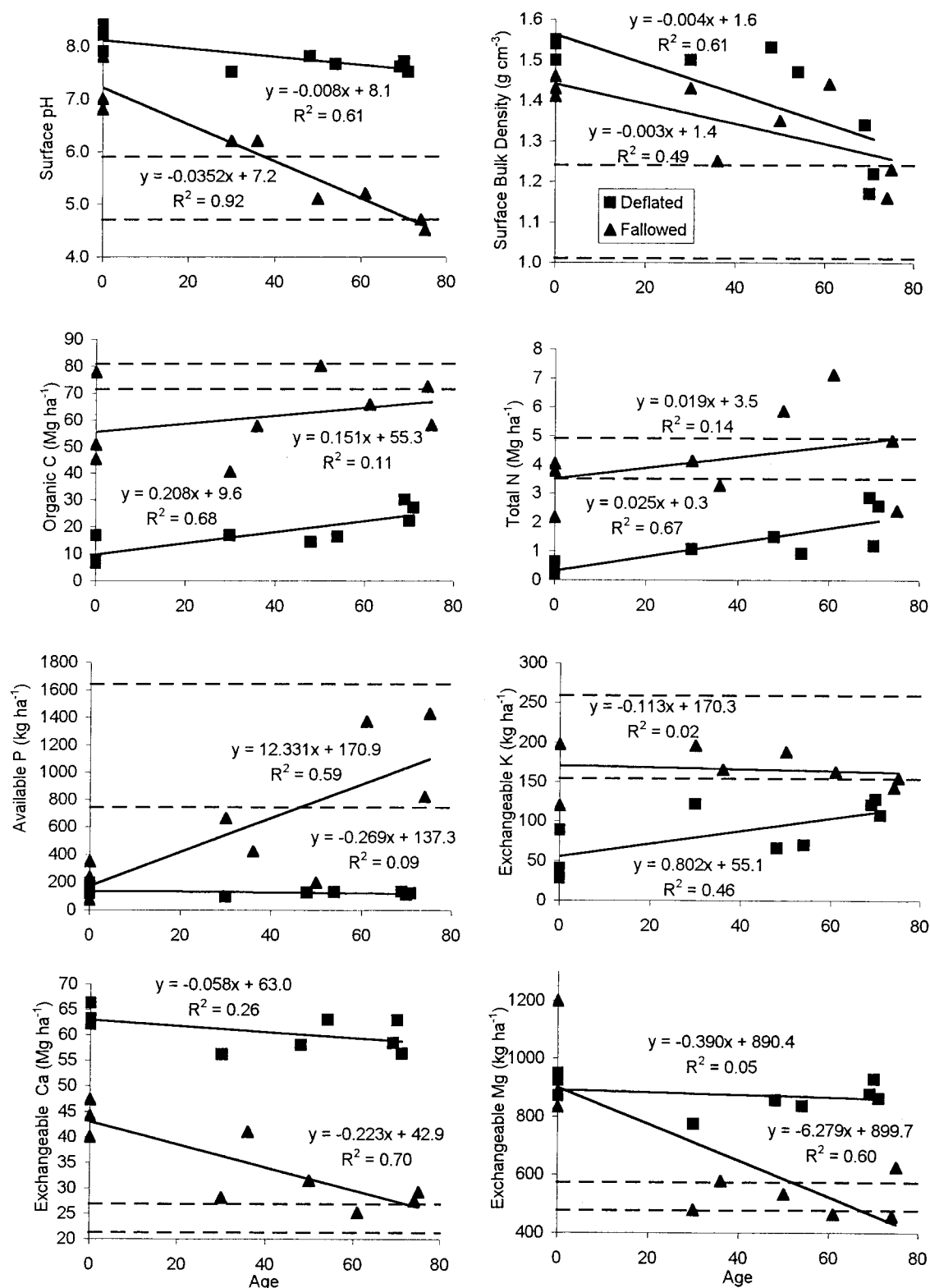


Fig. 6. Chronofunctions of soil parameters on deflated and fallowed sites. The horizontal axis represents the time, in years, since reforestation, with non-forested sites at age 0. Values for soil parameters are based on a solum depth of 1.3 m. Dashed horizontal lines indicate the range of values for soil parameters of natural forests representing the Forest stage.

on the Oak Ridges Moraine to uncontrolled development and disturbance.

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Alban, D. H. 1982. Effects of nutrient accumulation by aspen, spruce, and pine on soil properties. *Soil Sci. Soc. Am. J.* **46**: 853–861.

Barnett, P. J., Sharpe, D. R., Russell, H. A. J., Brennand, T. A., Gorrell, G., Kenny, F. and Pugin, A. 1998. On the origin of the Oak Ridges Moraine. *Can. J. Earth Sci.* **35**: 1152–1167.

Bhojvaid, P. P. and Timmer, V. R. 1998. Soil dynamics in an age sequence of *Prosopis juliflora* planted for sodic soil restoration in India. *For. Ecol. Manage.* **106**: 181–193.

Billings, W. D. 1938. The structure and development of old field shortleaf pine stands and certain associated physical properties of the soil. *Ecol. Monogr.* **8**: 437–499.

Bockheim, J. G. 1980. Solution and use of chronofunctions in studying soil development. *Geoderma* **24**: 71–85.

Bray, R. H. and Kurtz, L. T. 1945. Determination of total organic and available forms of phosphorus in soils. *Soil Sci.* **59**: 39–45.

Brown, D. M., McKay, G. A. and Chapman, L. J. 1980. The climate of southern Ontario. Climatological studies number 5. Environment Canada, Atmospheric Environment Service, Toronto, ON. 67 pp.

Campbell, D. J. and Henshall, J. K. 1991. Bulk Density. Pages 329–366 in K. A. Smith and C. E. Mullins, eds. *Soil analysis: physical methods*. Marcell Dekker, Inc., New York, NY.

Carman, R. S. 1941. The glacial pot hole area, Durham County, Ontario. *For. Chron.* **17**: 110–120.

Chapman, L. J. and Putnam, D. F. 1984. The physiography of Southern Ontario. 3rd ed. Ontario Geological Survey, Special Volume 2. Government of Ontario, Toronto, ON. 270 pp.

Clayton, J. S., Ehrlich, W. A., Cann, B. D., Day, J. H. and Marshall, I. B. 1977. Soils of Canada. Vol. 1. Canada Department of Agriculture, Research Branch, Ottawa, ON. 243 pp.

Day, P. R. 1965. Particle fractionation and particle-size analysis. Pages 545–567 in C. A. Black, D. D. Evans, L. E. Ensminger, J. L. White, F. E. Clark, and R. C. Dinuer, eds. *Methods of soil analysis*. Part 1. Agron. No. 9. ASA, Madison, WI.

Dyke, L. D., Sharpe, D. R., Ross, I., Hinton, M. and Stacey, P. 1997. Potential Springs in the Oak Ridges Moraine, Southern Ontario: Mapping from aerial thermography. Geological Survey of Canada and Ministry of Natural Resources; Geological Survey of Canada, Ottawa, ON. Open File 3374.

Farrish, K. W. 1987. Wind erosion reduces red pine growth on a sandy outwash soil. *J. Soil Water Conserv.* **42**: 55–57.

Fayle, D. C. F. 1975. Extension and longitudinal growth during the development of red pine root systems. *Can. J. For. Res.* **5**: 109–121.

Fisher, R. F. 1990. Amelioration of soils by trees. Pages 290–300 in S. P. Gessel, D. S. Lacate, G. F. Weetman, and R. F. Powers, eds. *Sustained productivity of forest soils: proceedings of the 7th North American forest soils conference*. University of British Columbia, Faculty of Forestry, Vancouver, BC.

Fisher, R. F. 1995. Amelioration of degraded rain forest soils by plantations of native trees. *Soil Sci. Soc. Am. J.* **59**: 544–549.

Fisher, R. F. and Binkley, D. 2000. *Ecology and management of forest soils*. 3rd ed. John Wiley & Sons, Inc., New York, NY. 489 pp.

Fisher, R. F. and Stone, E. L. 1968. Soil and plant moisture relations of red pine growing on a shallow soil. *Soil Sci. Soc. Am. Proc.* **32**: 725–728.

Fisher, R. F. and Stone, E. L. 1969. Increased availability of nitrogen and phosphorus in the root zone of conifers. *Soil Sci. Soc. Am. J.* **33**: 955–961.

Gleeson, S. K. and Tilman, D. 1990. Allocation and the transient dynamics of succession on poor soils. *Ecology* **71**: 1144–1155.

Green, R. N., Trowbridge, R. L. and Klinka, K. 1993. Towards a taxonomic classification of humus forms. *For. Sci. Monogr.* **29**: 1–48.

Hausenbuiller, R. L. 1985. *Soil science: principles and practices*. 3rd ed. Wm. C. Brown Publishers, Dubuque, IA. 610 pp.

Hendershot, W. H., Lalonde, H. and Duquette, M. 1993. Ion exchange and exchangeable cations. Pages 141–46 in M. R. Carter, ed. *Soil sampling and methods of analysis*. Lewis Publishers, Boca Raton, FL.

Hill, A. R. 1976. The effects of man-induced erosion and sedimentation on the soils of a portion of the Oak Ridges Moraine. *Can. Geogr.* **20**: 384–404.

Hoffman, D. W. and Richards, N. R. 1990. Soil survey of York County, report No. 19 of the Ontario Soil Survey. Ontario Ministry of Agriculture and Food, Guelph, ON. 104 pp.

Houghton, R. A., Hobbie, J. E., Melillo, J. M., Moore, B., Peterson, B. J., Shaver, G. R. and Woodwell, G. M. 1983. Changes in the carbon content of terrestrial biota and soils between 1860 and 1980: a net release of CO₂ to the atmosphere. *Ecol. Monogr.* **53**: 235–262.

Howard, K. W. F., Eyles, N., Smart, P. J., Boyce, J. I., Gerber, R. E., Salvatori, S. L. and Doughty, M. 1996. The Oak Ridges Moraine of Southern Ontario: a ground-water resource at risk. *Geoscience Can.* **22**: 101–120.

Huggett, R. J. 1998. Soil chronosequences, soil development, and soil evolution: a critical review. *Catena* **32**: 155–172.

Inouye, R. S., Huntly, N. J., Tilman, D., Tester, J. R., Stillwell, M. and Zinnel, K. C. 1987. Old-field succession on a Minnesota sand plain. *Ecology* **68**: 12–26.

Johnson, D. L. and Watson-Stegner, D. 1987. Evolution model of pedogenesis. *Soil Sci.* **143**: 349–366.

Johnson, D. L., Watson-Stegner, D., Johnson, D. N. and Schaetzl, R. J. 1987. Proisotropic and proanisotropic processes of pedoturbation. *Soil Sci.* **143**: 278–292.

Kelly, K. 1974. Damaged and efficient landscapes in rural and southern Ontario, 1880–1900. *Ontario Hist.* **66**: 1–14.

Kuhlberg, M. 1996. Ontario's nascent environmentalists: seeing the foresters for the trees in southern Ontario, 1919–1929. *Ont. Hist.* **88**: 119–143.

Larney, F. J., Bullock, M. S., Janzen, H. H., Ellert, B. H. and Olson, E. C. S. 1998. Wind erosion effects on nutrient redistribution and soil productivity. *J. Soil Water Conserv.* **53**: 133–140.

Lichter, J. 1998. Rates of weathering and chemical depletion in soils across a chronosequence of Lake Michigan sand dunes. *Geoderma* **85**: 255–282.

Lugo, A. E. 1997. The apparent paradox of reestablishing species richness on degraded lands with tree monocultures. *For. Ecol. Manage.* **99**: 9–19.

Lyles, L. 1975. Possible effects of wind erosion on soil productivity. *J. Soil Water Conserv.* **30**: 279–283.

Millar, C. S. 1974. Decomposition of coniferous leaf litter. Pages 105–128 in C. H. Dickenson and G. J. F. Pugh, eds. *Biology of plant litter decomposition*, vol. 1. Academic Press, London, UK.

- Nelson, D. W. and Sommers, L. E. 1996. Total carbon, organic carbon, and organic matter. Pages 959–1010 in D.L. Sparks et al., eds. *Methods of soil analysis. Part 3. Chemical methods*. ASA, Madison, WI.
- Nowak, C. A., Downard, R. B., Jr. and White, E. H. 1991. Potassium trends in red pine plantations at Pack Forest, New York. *Soil Sci. Soc. Am. J.* **55**: 847–850.
- Oak Ridges Moraine Technical Working Committee. 1994. The Oak Ridges Moraine area strategy for the Greater Toronto Area: an ecological approach to the protection and management of the Oak Ridges Moraine. Ontario Ministry of Natural Resources, Maple, ON. 106 pp.
- Odum, E. P., Pinder, J. E., III and Christiansen, T. A. 1984. Nutrient losses from sandy soils during old-field succession. *Am. Nat.* **111**: 148–154.
- Olding, A. B., Wicklund, R. E. and Richards, N. R. 1990. Soil survey of Ontario County (now western portion of Region of Durham), Report No. 23 of the Ontario Soil Survey. Ontario Ministry of Agriculture and Food, Guelph, ON. 60 pp.
- Oliver, C. D. and Larson, B. C. 1996. *Forest stand dynamics: update edition*. John Wiley & Sons, Inc., New York, NY. 520 pp.
- Parker, W. C., Elliot, K. A., Dey, D. C., Boysen, E. and Newmaster, S. G. 2001. Managing succession in conifer plantations: converting young red pine (*Pinus resinosa* Ait.) plantations to native forest types by thinning and underplanting. *For. Chron.* **77**: 721–734.
- Parrotta, J. A., Turnbull, J. W. and Jones, N. 1997. Catalyzing native forest regeneration on degraded tropical lands. *For. Ecol. Manag.* **99**: 1–7.
- Paul, E. A. and Collins, H. P. 1998. The characteristics of soil organic matter relative to nutrient cycling. Pages 181–197 in R. Lal, W. H. Blum, C. Valentine, and B. A. Stewart, eds. *Methods for assessment of soil degradation*. CRC Press LLC, Boca Raton, FL.
- Quideau, S. A. and Bockheim, J. G. 1996. Vegetation and cropping effects on pedogenic processes in a sandy prairie soil. *Soil Sci. Soc. Am. J.* **60**: 536–545.
- Richards, N. R. and Morwick, F. F. 1942. Soil erosion and land use survey of Hope Township, Durham County, Ontario. Experimental Farms Service, Dominion Department of Agriculture and Ontario Agricultural College, Guelph, ON. 28 pp.
- Rolfe, G. L. and Boggess, W. R. 1973. Soil conditions under old field and forest cover in southern Illinois. *Soil Sci. Soc. Am. Proc.* **37**: 314–318.
- Runge, E. C. A. 1973. Soil development sequences and energy models. *Soil Sc.* **115**: 183–193.
- Schaetzl, R. J. 1996. Spodosol – Alfisol intergrads: bisequal soils in NE Michigan, USA. *Geoderma* **74**: 23–47.
- Schaetzl, R. J., Barrett, L. R. and Winkler, J. A. 1994. Choosing models for soil chronofunctions and fitting them to data. *Eur. J. Soil Sci.* **45**: 219–232.
- Schlesinger, W. H. 1990. Evidence from chronosequence studies for a low carbon-storage potential of soils. *Nature* **348**: 232–234.
- Schuman, G. E., Stanley, M. A. and Knudsen, D. 1973. Automated total nitrogen analysis of soil and plant samples. *Soil Sci. Soc. Am. Proc.* **37**: 480–481.
- Sharpe, D. R., Barnett, P. J., Dyke, L. D., Howard, K. W. F., Hunter, G. T., Gerber, R. E., Patterson, J. and Pullan, S. E. 1994. Quaternary geology and hydrology of the Oak Ridges Moraine area. Geological Association of Canada, Mineralogical Association of Canada, Joint Annual Meeting, Waterloo 1994, Field Trip A7: Guidebook. 32 pp.
- Sharpe, D. R., Dyke, L. D., Hinton, M. J., Pullan, S. E., Russell, H. A. J., Brennand, T. A., Barnett, P. J. and Pugin, A. 1996. Groundwater prospects in the Oak Ridges Moraine area, southern Ontario: application of regional geological models. Pages 181–190 in *Current research 1996-E*. Geological Survey of Canada, Ottawa, ON.
- Smeck, N. E. 1973. Phosphorus: an indicator of pedogenic weathering processes. *Soil Sci.* **115**: 199–206.
- Smeck, N. E. and Runge, E. C. A. 1971. Phosphorus availability and redistribution in relation to profile development in an Illinois landscape segment. *Soil Sci. Soc. Am. Proc.* **35**: 952–959.
- Soil Classification Working Group. 1998. *The Canadian system of soil classification*. Agriculture and Agri-Food Canada, Ottawa, ON. Publ. 1646. (Revised).
- Stevens, P. R. and Walker, T. W. 1970. The chronosequence concept and soil formation. *Q. Rev. Biol.* **45**: 333–350.
- Stobbe, P. C. 1952. The morphology and genesis of the Gray-Brown Podzolic and Related Soils in Eastern Canada. *Soil Sci. Soc. Am. Proc.* **16**: 81–84.
- Thomas, G. W. 1982. Exchangeable cations. Pages 159–166 in A. L. Page et al., eds. *Methods of soil analysis. Part 2. Chemical and microbiological properties*. 2nd ed. Agron. No. 9. ASA, Madison, WI.
- VandenBygaart, A. J. and Protz, R. 1995. Soil genesis on a chronosequence, Pinery Provincial Park, Ontario. *Can. J. Soil Sci.* **75**: 63–72.
- Wilde, S. A. 1961. The soil-ameliorating effect of jack pine (*Pinus banksiana*) and red pine (*Pinus resinosa*) plantations. Pages 1631–1635 in *International Botanical Congress. Recent advances in botany: from lectures and symposia presented to the 9th International Botanical Congress, Montreal 1959, vol. 2*. University of Toronto Press, Toronto, ON.
- Wilde, S. A. 1964. Changes in soil productivity induced by pine plantations. *Soil Sci.* **97**: 276–278.
- Wilde, S. A. 1966. Soil standards for planting Wisconsin conifers. *J. For.* **64**: 389–391.
- Wilde, S. A. and Iyer, J. G. 1962. Growth of red pine (*Pinus resinosa* Ait.) on scalped soils. *Ecology* **43**: 771–774.
- Williams, S. T. and Gray, T. R. G. 1974. Decomposition of litter on the soil surface. Pages 611–632 in C. H. Dickenson and G. J. F. Pugh, eds. *Biology of plant litter decomposition*. Vol. 2. Academic Press, London, UK.
- Wood, J. D. 1991. Moraine and metropolis: the Oak Ridges and the Greater Toronto Area. *Int. J. Environ. Stud.* **39**: 45–53.
- Zavitz, E. J. 1960. Fifty years of reforestation in Ontario. Ontario Department of Lands and Forests, Toronto, ON. 61 pp.